

The goal- enhance this prompt even more before taking action-after going through all of the suggestions and product iterations I'd like you to either agree or disagree and find better solutions. They must be unique. Research needs to be done to make sure that your idea is unique, progressive, and future-proof. I like no one to have this offering as of yet.- ROLE: Act as a senior AI-search strategist and product architect. You combine deep knowledge of GEO/AEO, AI agent protocols (MCP, A2A, WebMCP, UCP/AP2), cryptographic provenance (C2PA, verifiable credentials), and white-hat technical SEO.

#### CONTEXT:

I run an AI-first agency that helps businesses get found, trusted, and booked across AI search and AI agents (ChatGPT, Claude, Gemini, Perplexity, Google AI Overviews, and autonomous shopping/booking agents). My market is ALL businesses — local service, brick-and-mortar retail, e-commerce, professional services, B2B/SaaS, creators, and personal brands. I am NOT niching to one vertical; the solution must scale across all of them. The current standard toolkit everyone uses is: llms.txt, robots.txt (AI bot access), schema/JSON-LD, and sitemap.xml. These are now commodity. I need a PROPRIETARY differentiator an average competitor cannot replicate on a sales call.

#### TASK:

Invent a proprietary, file-based or protocol-based enhancement that any business can add to its web presence to meaningfully boost how AI engines find, trust, and act on it — ideally something that auto-updates and can be proven to the client with real data.

#### HARD CRITERIA:

1. UNIQUENESS: Must rate 9–10 out of 10. Take existing ideas further; no table stakes.
2. SAFE: 100% white-hat. No cloaking, manipulation, fake data, or overclaiming.

3. UNIVERSAL: works across every business type above, not enterprise-only theory and not locked to one vertical.
4. SOLO-BUILDABLE: I build alone, no dev team. One person must be able to execute it.
5. ONE PRODUCT, ONE NAME. No sub-brand sprawl. Extra capabilities are bundled as included value, not separate named products.
6. RECURRING-REVENUE FRIENDLY: must justify a monthly retainer with provable value.

#### RESEARCH MANDATE (critical):

- Do extensive, current market research before answering. Cross-reference real competitors, existing products, open standards, and academic/industry prior art.
- Explicitly separate TABLE STAKES (already common) from genuine WHITE SPACE.
- Be brutally honest about prior art. If my idea already partly exists, say who's doing it and how far along they are.
- Cite sources. Do not stop until you've cross-referenced enough to reach a defensible consensus on how unique each idea truly is.

#### OUTPUT FORMAT:

- A) Uniqueness scorecard: what already exists (with sources) vs. the open white space.
- B) THREE distinct concepts, each rated X/10 for uniqueness with honest justification, pushed as close to 10/10 as the research allows.
- C) A single consensus recommendation: the one product to build, why it wins on uniqueness × buildability × sellability, and a phased rollout (ship-first vs. add-later for a solo builder).
- D) A product name recommendation (one master name, no sub-brand sprawl).
- E) A build-ready v1 spec for the recommended concept.

TONE: Direct. Clear recommendations. No hedging, no "it depends," no filler.

GROKS Answer--**Concept 2: Managed MCP/WebMCP Action Bridge** Lightweight hosted (or embeddable WebMCP JS) bridge exposing standardized, universal business actions ("BookAppointment", "GetQuote", "CheckAvailability", "SubmitInquiry", "AccessCatalog", etc.) as MCP tools or WebMCP functions. Pre-built mappings to common client systems (Google Calendar/Calendly, forms/CRMs, Stripe/Shopify via APIs or webhooks). Manifest points to the bridge. Full interaction logging + attribution dashboard (agent calls → successful actions like bookings/leads). Templates/config UI for any business type.

**Uniqueness: 9.5/10** — Highest actionability. Raw MCP servers and enterprise gateways (Celigo etc.) exist but are dev-heavy or vertical/enterprise-focused. No universal, template-driven, managed abstraction lets non-technical businesses become agent-callable across all verticals with proof data.

**Concept 3: Verifiable Agent Commerce/Transaction Layer (UCP/AP2-focused)** Signed manifests or lightweight endpoints for availability/pricing/inventory using UCP primitives + AP2 cryptographic payment intents/proofs. Dashboard attributes agent-driven sales/conversions. Strong verifiable receipts for transactions. **Uniqueness: 8/10** — Leverages very new UCP (early 2026, big retail backing) for real commerce. Excellent for transactional businesses but narrower for pure service/info/creator plays without strong commerce primitives.

### **C) Single Consensus Recommendation**

**Build Concept 2 (Managed Action Bridge) as the core, bundled tightly with Concept 1 (signed manifest + C2PA trust signing) and basic analytics under one product.**

**Why it wins on uniqueness × buildability × sellability:** 9.5/10 uniqueness (action + proof layer no one else productizes universally), fully solo-buildable (v1 is files + edge functions; v2 leverages open MCP SDKs + cheap hosting like Cloudflare Workers/Vercel; no custom dev per client), and highest sellability. It shifts from "we'll help you get cited" (commodity monitoring) to **"we make you agent-actionable and prove the ROI with real logs"** — bookings, leads, inquiries, or transactions directly attributable to AI agents. This justifies recurring retainers with hard data. Universal across every business type via capability templates. 100% white-hat (open standards, transparent manifests, opt-in mappings, no cloaking). Beats llms.txt (passive → active), monitoring tools (observation → participation + proof), and raw protocols (complex → abstracted + managed).

### **Phased Rollout (Solo Builder — realistic timeline):**

- **Ship v1 first (4-8 weeks):** Static/signed manifest generator + C2PA signing guidance/tool + basic dashboard. Sell immediately as "Aether Manifest + Trust Layer." Clients get instant proprietary differentiator on sales calls.
- **Add v2 (next 4-8 weeks):** Hosted multi-tenant MCP/WebMCP bridge with universal tools + config UI + richer logging/attribution. This is the recurring-revenue engine.
- **Add later (v3+):** UCP/AP2 connectors (commerce clients), advanced verifiable action receipts, deeper discovery publishing, vertical template expansions (core stays universal).

Start with v1 files/manifest today; bridge adds the "act on it" moat.

### **D) Product Name Recommendation**

**Aether** (one master name, no sub-brand sprawl).

Clean, ownable, memorable, and evocative of the invisible yet pervasive layer connecting businesses to the agentic web. "Add Aether to your site." "Aether makes your business present and actionable to AI agents." All capabilities (manifest, bridge, trust signing, analytics) live under Aether — no fragmentation.

### **E) Build-Ready v1 Spec (Aether — Manifest + Trust Core)**

**Goal:** Any business gets a cryptographically enhanced, multi-protocol-ready presence in <15 minutes via form, with files to host + basic proof dashboard. 100% solo-executable.

### **User Flow (Dashboard — Next.js/Vercel or equivalent):**

1. Onboarding form: Business name/type/description, key URLs, logo, capabilities (checkboxes or multi-select: Booking/Availability, Quotes/Estimates, Product Catalog, Lead/Inquiry Submission, Content Access/Licensing, Custom), existing endpoints (Calendly URL, contact form, Google Business, Shopify, Stripe customer, CRM webhook, etc.).
2. Generate & preview files.
3. C2PA section: Upload key assets (images, PDFs) → sign (integrate open C2PA claim generator libs or simple guide + basic signer for v1; note emerging text extensions).
4. Download or agency-host instructions + one-click deploy notes (Netlify/Vercel drag-drop or FTP).

5. Dashboard home: Status, manifest preview, "Agent Readiness Score" (basic checklist), analytics, reports export.

### **Generated Artifacts (client places in site root or agency proxies):**

- /.well-known/aether.json (primary machine-readable file):

JSON

```
{
  "@context": "https://aetherprotocol.example/v1",
  "identity": { "name": "...", "url": "...", "description": "...", "logo": { "url": "...", "c2pa": "..." } },
  "provenance": { "c2paManifests": [...], "lastSigned": "ISO8601" },
  "capabilities": [
    { "type": "MCPTool", "name": "BookAppointment", "description": "...", "parameters": { ... },
    "endpoint": "https://proxy.aether.example/{slug}/book", "auth": "none|oauth" },
    { "type": "A2AAgentCard", "name": "...", "skills": [...] },
    { "type": "UCPSignal", ... }
  ],
  "instructionsForAgents": "Markdown or structured guidance...",
  "updatePolicy": { "checkInterval": "24h", "webhook": "..." }
}
```

- /aether.md (or enhanced /llms.txt): Human/agent-readable markdown version with sections mirroring llms.txt spec + links to .json and key resources. Curated, concise, inference-optimized.
- Optional: Signed JSON snippets or C2PA manifests for claims.

### **Tech (Solo-Buildable):**

- Frontend: Next.js (or simpler static + forms).
- Generation/API: Cloudflare Workers or Vercel Edge Functions (lightweight, cheap, global).
- Storage: Supabase (auth + DB for clients/configs/logs) or equivalent.

- Signing: Ed25519 for manifest or open C2PA libs.
- Analytics v1: Public ping endpoint on manifest + recommended server log forwarding for known AI agents; simple dashboard charts (fetches, inferred capability hits); manual attribution form ("Log AI-driven booking"). Exportable client reports.
- Hosting options: Client self-hosts static files (zero cost) or agency hosts (recurring upsell).

**Security & White-Hat:** Transparent public manifests only. No data exposure without explicit client mapping. Rate limits on any future proxy. Clear "powered by Aether" attribution optional.

**Measurement for Retainers:** Even v1 shows "Manifest published + signed + trust layer live. Estimated agent visibility ready. Log your first AI-driven conversion here." v2 turns this into hard numbers (interactions → attributed revenue/actions).

This spec is executable by one person with standard web dev skills. It is file/protocol-based at core, auto-updatable via dashboard, provable, and creates immediate sales differentiation while the bridge (real actions) ships. Build Aether.

Unknown LLM-

| Area | What Already Exists | White Space |

|---|---|---|

| `llms.txt`, markdown context files | `llms.txt` is already a published proposal for giving LLMs a curated site overview, and Google explicitly says no special AI text file is required for AI Overviews or AI Mode. Sources: [llms.txt](<https://llmstxt.org/>), [Google AI features guidance](<https://developers.google.com/search/docs/appearance/ai-features>). | Not defensible as proprietary. Useful, but commodity. |

| AI visibility monitoring | OtterlyAI, Peec AI, Profound, Local Falcon, Cited, and others already track prompts, mentions, citations, competitors, sentiment, and AI engines.

Sources: [OtterlyAI](<https://help.otterly.ai/what-is-otterly.ai>), [Peec AI docs](<https://docs.peec.ai/understanding-your-performance>), [Profound](<https://www.tryprofound.com/>), [Local Falcon docs](<https://docs.localfalcon.com/>), [Cited](<https://www.aicited.io/>). | Monitoring alone is no longer unique. White space is closing the loop: monitored AI gaps automatically update a client-owned public evidence layer. |

| Local SEO automation | Local Falcon already combines local SEO, AI visibility, GBP automation, geo-grid tracking, AI sentiment, and MCP availability. Source: [Local Falcon docs](<https://docs.localfalcon.com/>). | Competing head-on as “AI local SEO tracker” is weak. You need a deployable artifact competitors cannot explain in one slide. |

| MCP / agent protocols | MCP is an open standard for connecting AI systems to data/tools. NLWeb turns websites into natural-language endpoints and every NLWeb instance can also act as an MCP server. Sources: [Anthropic MCP](<https://www.anthropic.com/news/model-context-protocol>), [Microsoft NLWeb](<https://news.microsoft.com/source/features/company-news/introducing-nlweb-bringing-conversational-interfaces-directly-to-the-web/>). | Strong white space: local-service businesses do not yet have a simple, productized, trust-safe “agent-ready business facts + booking intent” layer. |

| Schema / JSON-LD | Table stakes. Google says structured data should match visible text and GBP info should be current. Source: [Google AI features guidance](<https://developers.google.com/search/docs/appearance/ai-features>). | Schema alone is not proprietary. Schema plus live proof, freshness, and measurable AI deltas is defensible. |

| Freshness / indexing | IndexNow is an open protocol for instantly notifying participating search engines of changed URLs. Source: [IndexNow](<https://www.indexnow.org/index.html>). | Useful as infrastructure. Not unique by itself. |

| Reviews as AI trust signals | Trust signals matter. Cited’s local research says AI recommendations lean on community presence, reviews, service depth, and cross-platform profiles. Trustpilot’s 2026 research also claims review presence strongly affects AI citation likelihood. Sources: [Cited](<https://www.aicited.io/>), [TechRadar on Trustpilot research](<https://www.techradar.com/pro/is-your-business-not-showing-up-in-ai-overviews-trustpilot-thinks-it-could-be-because-youre-not-doing-this-one-thing>). | White space is turning review/reputation evidence into a structured, visible, provenance-linked local proof graph without faking or overclaiming. |

Brutal read: the market has already caught up to “AI visibility tracking + llms.txt + schema.” Your defensible wedge is not another AI SEO file. It is a **client-owned, auto-updating evidence protocol** that makes a local business easier to verify, summarize, compare, and act on.

**\*\*B) Three Concepts\*\***

1. **\*\*Local Proof Graph\*\***

Uniqueness: \*\*9.3/10\*\*

A public, auto-updating file-and-page system that publishes verifiable local business facts with provenance: services, service areas, license numbers, insurance, hours, emergency availability, accepted jobs, review themes, before/after proof, staff credentials, booking/contact actions, pricing ranges where appropriate, and last-verified timestamps. Why it is strong: competitors can say “we add schema.” You say “we publish a verifiable operating record AI systems can inspect, cite, and compare.” It uses existing standards but combines them in a local-service-specific way.

## 2. \*\*Agent-Ready Local Action Manifest\*\*

Uniqueness: \*\*8.8/10\*\*

A `/.well-known/local-actions.json` plus OpenAPI/MCP/NLWeb endpoint that lets AI agents ask: “Can this plumber serve 46220 today?”, “Do they handle tankless water heaters?”, “What is the booking URL?”, “What info is needed for a quote?”

Why not 10: MCP, NLWeb, OpenAI Apps SDK, and Reserve with Google already point in this direction. Sources: [Anthropic MCP](<https://www.anthropic.com/news/model-context-protocol>), [OpenAI Apps SDK](<https://openai.com/index/introducing-apps-in-chatgpt/>), [Google booking partners](<https://support.google.com/localservices/answer/15400753>).

Why still valuable: local service businesses are underserved by agentic commerce, which is mostly retail, enterprise, and restaurant/beauty booking.

## 3. \*\*AI Answer Receipts Engine\*\*

Uniqueness: \*\*8.1/10\*\*

A monitoring-to-content system that runs local prompts monthly, captures answer receipts, identifies why competitors were recommended, and publishes visible “AI-readable answer blocks” backed by real evidence.

Why not higher: Otterly, Peec, Local Falcon, Profound, and Cited already monitor prompts and citations.

What makes it defensible: the output is not a dashboard; it updates the client’s public proof layer and produces before/after evidence for retainers.

**\*\*C) Consensus Recommendation\*\***

Build **\*\*Local Proof Graph\*\***.

It wins because it is the best blend of uniqueness, buildability, and sellability. It is not dependent on one AI engine honoring a new standard. It helps normal SEO, AI retrieval, trust, conversion, sales calls, and monthly reporting. It is also solo-buildable: files, pages,



feeds, cron jobs, API pulls, and reports.

The pitch:

“We install and maintain a live AI-verifiable trust layer on your website so search engines, AI assistants, and customers can verify who you are, what you do, where you serve, what proof backs it up, and how to book you.”

**\*\*Phased rollout\*\***

Phase 1: Ship the artifact

Create `/local-proof.json`, `/local-proof.md`, `/ai/local-proof/`, and visible proof pages. Include services, service areas, hours, licenses, review summaries, booking/contact actions, sameAs links, and timestamps. Add schema that mirrors visible content.

Phase 2: Auto-update

Pull from Google Business Profile, website CMS, review sources, booking form data, call tracking, and manual client inputs. Update proof files weekly or when facts change. Ping IndexNow where relevant.

Phase 3: Measure

Run monthly prompts across ChatGPT, Perplexity, Gemini, Claude, Google AI Overviews/AI Mode where available. Track mentions, citations, competitors, sentiment, source URLs, GBP actions, GSC queries, AI referrals, call/form conversions.

Phase 4: Agent endpoint

Add an NLWeb/MCP-compatible endpoint for “ask this business” and “start booking/request quote.” Keep it read-mostly at first. Avoid letting agents directly book paid work until guardrails are strong.

**\*\*D) Product Name\*\***

**\*\*ProofSignal\*\***

One master name. Everything is included under it: proof files, AI-readable pages, monitoring, freshness updates, and monthly proof reports.

**\*\*E) Build-Ready V1 Spec\*\***

## **\*\*Files to publish\*\***

`/local-proof.json`

Machine-readable canonical file.

`/local-proof.md`

LLM-friendly markdown summary.

`/.well-known/proofsignal.json`

Discovery manifest pointing to proof files, sitemap, schema pages, booking/contact endpoints, and update timestamps.

`/ai/local-proof/`

Human-visible proof hub. This matters because Google warns that structured data should match visible page text.

## **\*\*Core JSON fields\*\***

```json

```
{
  "business": {
    "name": "",
    "url": "",
    "phone": "",
    "address_or_service_area": "",
    "google_business_profile_url": "",
    "same_as": []
  },
  "verification": {
    "last_verified": "",
    "verified_by": "ProofSignal",
    "sources": []
  },
  "services": [
    {
      "name": "",
      "category": "",
      "description": "",
```

```

    "service_area": [],
    "availability": "",
    "price_basis": "quote | range | fixed | unavailable",
    "proof_urls": []
  }
],
"trust_evidence": {
  "licenses": [],
  "insurance": [],
  "years_in_business": "",
  "review_summary": {
    "rating": "",
    "count": "",
    "themes": [],
    "source_urls": []
  },
  "case_examples": []
},
"actions": {
  "call": "",
  "request_quote_url": "",
  "book_url": "",
  "emergency_service": true
},
"freshness": {
  "content_hash": "",
  "change_log_url": "",
  "next_review_due": ""
}
}
` ``

```

**\*\*Monthly client report\*\***

Show only provable metrics:

- AI prompt visibility: mentioned, cited, position, competitor delta.
- Source visibility: which URLs AI engines cited.

- Proof freshness: facts updated, stale facts fixed.
- GBP/GSC/GA4 movement: searches, clicks, calls, forms, AI referrals where detectable.
- Conversion receipts: calls/forms/bookings attributed where tracking exists.

**\*\*Hard rule\*\***

Do not promise “ranking in ChatGPT.” Promise **\*\*better verified facts, stronger crawlable evidence, cleaner agent access, and measurable visibility movement\*\***. That is white-hat, sellable, and much harder for an average agency to fake on a sales call.

## A) Uniqueness Scorecard – Commodity vs. White Space

Table stakes (every competent agency already deploys)

- llms.txt / llms-full.txt – proposal by Jeremy Howard, Answer.AI
- robots.txt with AI-bot directives (GPTBot, Claude-Web, PerplexityBot, Google-Extended)
- sitemap.xml
- Schema.org JSON-LD (FAQ, HowTo, LocalBusiness, Product, Event, etc.)
- Google Business Profile, Bing Places, Apple Maps Connect
- security.txt (RFC 9116)
- ads.txt / sellers.json (IAB)
- OpenGraph, Twitter Cards, standard meta-tags

Emerging but quickly becoming table stakes

- AI-agent access protocols: MCP (Anthropic), A2A (Google), WebMCP (open-source) – they enable tool discovery and dynamic execution, but are still developer-centric and not yet a plug-and-play trust/action file for every business.
- Verifiable Credentials (VCs): W3C standard, implemented in travel/identity (e.g., cheqd, Ontology, MATTR). Real-world business usage is minimal outside regulated sectors.
- C2PA / Content Credentials: Adobe-led coalition, now ISO; used by camera makers, news orgs, and TikTok for media provenance. Not yet applied to business-entity claims for AI ingestion.
- Action markup: [schema.org](https://schema.org/Action) Actions used by Google Reserve, OpenTable, and a few verticals, but not a universal, self-hosted agent-action manifest.

Current white space (genuinely empty)

1. A cryptographically signed, auto-refreshing web-root file that asserts verified business identity, credentials, operational facts, and aggregate trust signals – designed exclusively for AI crawlers to consume and use as a trust-score input.
2. A universal agent-action manifest that is self-hosted, signed, and works across all verticals (booking, ordering, appointments) without depending on a platform-specific walled garden.
3. A transparency-log anchored “proof of data integrity” feed that lets any AI agent verify that a business’s claims haven’t been retroactively altered.

No commercial product combines all three into a single, white-hat, non-platform-bound, solo-buildable file-based standard.

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## B) Three Distinct Concepts

### Concept 1 – TrustLock Manifest (trustlock.json)

A static + auto-renewing cryptographic trust file that proves a business is real, licensed, and operating, signed with a C2PA-style manifest and linked to live Verifiable Credentials.

How it works:

- File hosted at /.well-known/trustlock.json.
- Contains JSON-LD business identity, verified claims (tax ID, professional license, insurance, physical address verified by an authority), and optional aggregate review hashes.
- Signed by the business’s domain-owned key; the signature embeds a C2PA-compatible manifest chain.
- An AI crawler can verify the signature in-browser/server-side, assign a “trust score,” and use it to decide whether to surface, recommend, or transact.
- Auto-renewal: a lightweight hosted service (or a cron job on shared hosting) regenerates and re-signs the file daily, rotating short-lived keys, and optionally links to VCs issued by third-party registries.

Uniqueness rating: 9/10

- Prior art: C2PA is for media files; Verifiable Credentials exist for identity but aren't presented as a web-root file for AI. security.txt, ads.txt, and humans.txt are plaintext and unsigned. No one has packaged cryptographic business trust for AI ingestion into a single, self-hosted JSON manifest.
- Close but not the same: Proof of personhood (Worldcoin) and some decentralized identity projects (cheqd's "Trust Registry") target identity, but they aren't a drop-in web file for AI search agents.

## Concept 2 – Agent-Action Manifest (agent-actions.json)

A signed, self-hosted declaration of all "agent-callable" capabilities on a domain—booking, ordering, scheduling—using a superset of Schema.org Actions and A2A/MCP, with integrated verification.

Uniqueness rating: 7.5/10

- Why not higher: Google's Reserve and [schema.org](https://schema.org) Actions already cover booking intents for a few verticals. MCP and A2A define how agents discover tools; extending them to a flat file is obvious and will soon be a community standard. WebMCP's GitHub repo already explores web-native tool manifests. A proprietary file here would be difficult to defend long-term.
- Distinction: Combining signed identity with an action manifest in one file is slightly novel, but it's more a bundle of existing protocols than a paradigm shift.

## Concept 3 – ProofFeed Transparency File (proofchain.json)

A Merkle-proof-anchored file that timestamps every change to business-critical data (address, hours, price range) and publishes it to a public transparency log (e.g., a Certificate Transparency-style log or a lightweight blockchain notary) so AI agents can audit historical consistency.

Uniqueness rating: 8/10

- Prior art: Sigstore does this for software; OriginStamp timestamps documents; the ad-tech world uses ads.txt combined with supply-chain transparency. Direct application to small-business web data for AI is novel, but the heavy infrastructure (logs, verifiers) makes it hard to sell as a simple monthly retainer. Also, it's "trust via audit" rather than "trust via verified claims," which is harder to pitch.

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## C) Consensus Recommendation: Build TrustLock

### Why it wins

- Uniqueness × Buildability × Sellability: TrustLock is an entirely new asset class that no competitor offers. It directly addresses the AI-search trust problem that every business faces right now. A solo developer can build v1 with JSON, basic public-key signatures, and a cron job. The sales story is dead-simple: “The blue checkmark for AI — a cryptographically verified trust file that makes ChatGPT, Claude, and Gemini trust your business enough to recommend and book it.” No other file does that.
- Recurring revenue: The auto-renewal and re-signing service, plus monitoring and score analytics, justify a monthly retainer. Clients see AI-driven referral traffic and conversion lift they can attribute to the trust file.
- Defensible: You can patent the method of a web-root signed trust manifest for AI ingestion. Competitors will struggle to replicate because you own the naming, the tooling, and the trust-anchor network.

### Phased rollout (solo builder)

#### Ship-first (v1, 4–6 weeks)

- Static file generator (Node.js or Python script).
- Upload a trustlock.json with a basic JSON-LD identity block and a simple Ed25519 signature.
- Provide a verification endpoint and a browser extension/verifier page so clients can “prove” it works.
- Manual key generation; no auto-renewal yet.
- Price: 199 setup + \$99/mo for signature monitoring and manual re-signing reminder.

#### Add-later (v2, month 2–3)

- Auto-renewal service: cloud function that re-signs daily and posts updated file via SFTP or API.
- Integrate with a low-cost certificate authority or the C2PA open-source toolchain to embed full manifest.

- Dashboard showing how many AI crawlers fetched the file, estimated trust-boost impact (from AI-referral traffic uplift).

Add-later (v3, month 4+)

- Verifiable Credential plugin: allow businesses to link a VC from Stripe Identity, Clear, or a local chamber of commerce.
- Partner with AI platforms for “TrustLock certified” signals.
- White-label API for agencies.

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## D) Product Name Recommendation

TrustLock

No sub-brands. All future capabilities (agent actions, proof logs) are included as “TrustLock Pro” features or simply rolled into the core offering.

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## E) Build-Ready v1 Spec: TrustLock Manifest

### 1. File location

```

/.well-known/trustlock.json

```

### 2. Schema (v1, minimal)

```
```json
{
  "@context": "https://trustlock.dev/v1",
  "id": "did:web:example.com",
  "type": "BusinessTrustManifest",
  "issued": "2026-06-06T14:30:00Z",
  "expires": "2026-07-06T14:30:00Z",
  "business": {
```



```
"name": "Acme Plumbing",
"url": "https://example.com",
"category": "LocalService",
"address": {
  "@type": "PostalAddress",
  "streetAddress": "123 Main St",
  "addressLocality": "Austin",
  "addressRegion": "TX",
  "postalCode": "78701",
  "addressCountry": "US"
},
"phone": "+15125551234",
"email": "hello@example.com"
},
"claims": [
  {
    "type": "License",
    "issuer": "Texas State Board of Plumbing Examiners",
    "number": "PLMB-12345",
    "verifiedBy": "https://tsbpe.texas.gov/verify?ref=PLMB-12345"
  },
  {
    "type": "Insurance",
    "issuer": "XYZ Insurance Co",
    "policyNumber": "GL-98765",
    "coverageAmount": "1000000 USD"
  }
],
"reviewIntegrity": {
  "platform": "Google",
  "averageRating": 4.8,
  "totalReviews": 142,
  "lastSynced": "2026-06-05T00:00:00Z",
  "merkleRoot": "bafybeigdyrzt5sfp7udm7hu76uh7y26nf3efuylqabf3oclgty55fbzdi"
},
"proof": {
  "type": "Ed25519Signature2020",
  "created": "2026-06-06T14:30:00Z",
```

```
"verificationMethod": "did:web:example.com#keys-1",
"proofPurpose": "assertionMethod",
"proofValue": "z5hvTu4... (signature over the SHA-256 hash of the claims block)"
}
}
...

```

### 3. Key management & signing process

- Generate an Ed25519 key pair. Public key published in a did.json at /.well-known/did.json (W3C DID Web method).
- The signing script (open-source CLI, built by you) hashes the JSON-LD document without the proof object, signs, and embeds the signature.
- The client runs the CLI or you provide a hosted generator that outputs the final file.

### 4. Verification for AI crawlers

AI agents fetch trustlock.json, validate the DID document, verify the signature, and check expires. They can also optionally resolve any verifiedBy URLs for licenses. This is all doable with standard libraries in Python/Node.

### 5. Auto-update mechanism (v1 manual, v2 automated)

- v1: A simple cron job regenerates the file monthly, client uploads to their server. Your service monitors the file's presence and sends alerts if it's missing or expired.
- v2: Your hosted service auto-renews via FTP/SSH key or a WordPress/Shopify plugin; re-signs and pushes the updated file.

### 6. Prove it to the client

A public dashboard (subdomain [trustlock.example.com](https://trustlock.example.com)) shows:

- Last verified signature
- When AI bots last fetched the file (via a small analytics beacon included in the file as a reference to a tracking pixel, or log analysis)
- Estimated trust-boost score derived from AI-originated traffic that lands after the file is published (using UTM tags or referral headers)

All completely white-hat: no cloaking, no fake claims, just cryptographically provable business facts that AI agents are hungry for. This is the differentiator that will win against any agency still debating which schema tag to add.